# Доп домашнее задание

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

Все кодовые слова

000000
101101
001110
010111
100011
111010
011001
110100

$$G H^T = \emptyset$$

$x_1 + x_3 + x_4 + x_6 = 0$

$x_3 + x_4 + x_5 = 0$

$x_2 + x_4 + x_5 + x_6 = 0$

$x_3 = x_4 + x_5$

$x_1 = x_5 + x_6$

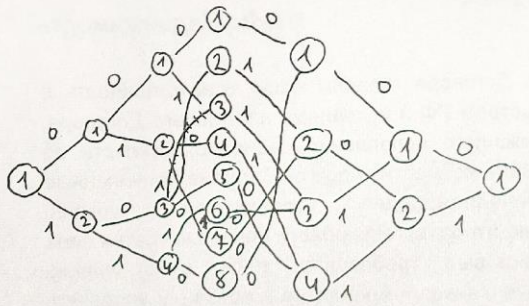
$x_2 = x_4 + x_5 + x_6$

| $x_5 + x_6$ | $x_4 + x_5 + x_6$ | $x_4 + x_5$ | $x_4$ | $x_5$ | $x_6$ |
|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 | 1 | 0 |

$$H = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 \end{pmatrix}$$

1) по кодовым словам

| $i$ | $\{c^p\}$ | $F_i\{c^p\}$ | $\{v_{i-1}\}$ | $v_i$ | $c_i$ |
|---|---|---|---|---|---|
| 0 | $\{\emptyset\}$ | $\{c_m\}$ | $\emptyset$ | 1 | — |
| 1 | 0<br>1 | 000000 1110<br>101 11, 11001<br>011 01, 00011<br>1101 0, 1010 0 | 1 | 1<br>2 | 0<br>1 |
| 2 | 00<br>01<br>10<br>11 | 0000, 1110<br>0111, 1001<br>1101, 0011<br>1010, 0100 | 1<br>1<br>2<br>2 | 1<br>2<br>3<br>4 | 0<br>1<br>0<br>1 |
| 3 | 000, 011<br>101, 110<br>001<br>010<br>100<br>111 | 000, 001<br>101, 100<br>110<br>111<br>011<br>010 | 1, 2<br>3, 4<br>1<br>2<br>3<br>4 | 1, 7<br>2, 8<br>3<br>4<br>5<br>6 | 0, 1<br>1, 0<br>1<br>0<br>0<br>1 |
| 4 | 0000, 1101<br>1011, 0110<br>0011, 1110<br>0101, 1000 | 00<br>01<br>10<br>11 | 1, 8<br>2, 7<br>3, 6<br>4, 5 | 1<br>2<br>3<br>4 | 0, 1<br>1, 0<br>1, 0<br>1, 0 |
| 5 | 00000, 00111<br>11101, 11010<br>10110, 01011<br>10001, 01100 | 0<br>1 | 1, 3, 3, 1<br>2, 4, 4, 2 | 1<br>2 | 0, 1, 1, 0<br>0, 1, 1, 0 |



2) по порождающей матрице

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix} \begin{matrix} -g_1 \\ -g_2 \\ -g_3 \end{matrix}$$

$g_1 = g_1 + g_3$

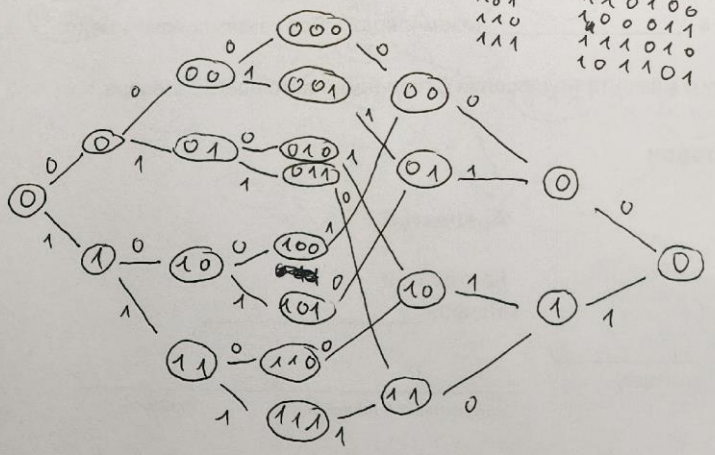
$$\begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$g_1 = g_1 + g_2$

$$\begin{matrix} m_1 \\ m_2 \\ m_3 \end{matrix} \begin{pmatrix} \boxed{110}100 \\ 00\boxed{110}0 \\ 0\boxed{1011}1 \end{pmatrix}$$

| m | 1 | 2 | 3 |
|---|---|---|---|
| | 0 | 0 | 0 |
| | 0 | 0 | 1 |
| | 0 | 1 | 0 |
| | 0 | 1 | 1 |
| | 1 | 0 | 0 |
| | 1 | 0 | 1 |
| | 1 | 1 | 0 |
| | 1 | 1 | 1 |

| | | | | | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 1 | 0 | 1 |



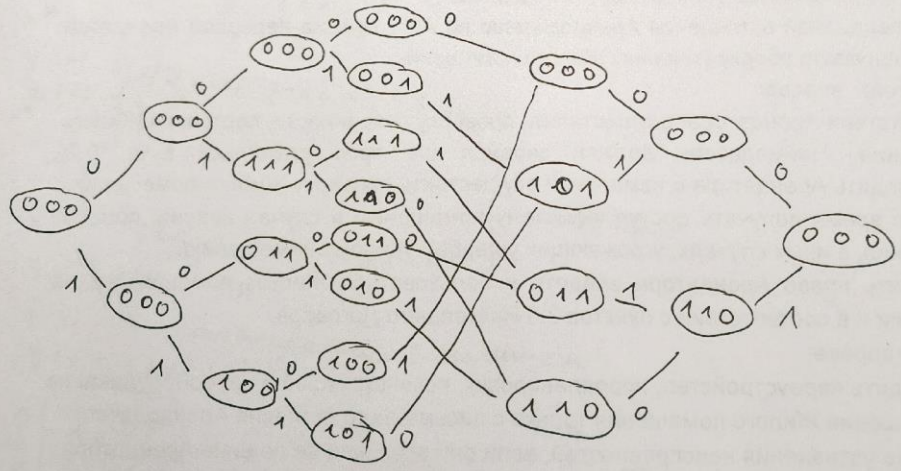
3) По проверочной матрице

$$H = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 \end{pmatrix}$$

| | | | | | |
|---|---|---|---|---|---|
| 0 | 000 | 0 | 000 | 000 | 000 |
| | | 1 | 011 | 01 | 111 |
| | | | | 10 | 011 |
| | | | | 11 | 100 |

| | |
|---|---|
| 000 | 000 |
| 001 | 001 |
| 010 | 111 |
| 011 | 110 |
| 100 | 011 |
| 101 | 010 |
| 110 | 101 |
| 111 | 101 |

| | |
|---|---|
| 0000 | 000 |
| 1011 | 110 |
| 0011 | 101 |
| 0101 | 011 |
| 1000 | 011 |
| 1110 | 101 |
| 0110 | 110 |
| 1101 | 000 |

| | |
|---|---|
| 00000 | 000 |
| 10110 | 000 |
| 00111 | 000 |
| 01011 | 110 |
| 10001 | 110 |
| 11101 | 000 |
| 01100 | 110 |
| 11010 | 110 |

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